Theorem 17: inapproximability for the operator p-norms

Correspondence between the paper and the Lean 4/Mathlib formalisation

Paper. J. Breitung, T. Gervens, M. Grohe. *Graph Similarity Meets Optimal Transport* (file `Graph_Similarity_Meets_Optimal_Transport.pdf` in the repository). Statement, equation, page and line numbers refer to this PDF; `docs/NUMBERING.md` lists the Lean files and declarations of all numbered statements.

The folder `GsimOT/Theorem17_OperatorNormHardness` formalises Eq. (45), Lemma 53, Corollary 54 and the reduction of Theorem 17 (p. 12, proof in Appendix G, pp. 54–57) from Three-Partition with Claims 55 and 56 and Eq. (46). It gives Theorem 4 (2). Shared notions come from `GsimOT/Common`; they are listed at the end.

Checking. Run `bash build.sh` in the repository root (see `README.md`). The script builds the library and runs `scripts/CheckAxioms.lean`, which must report only the standard axioms `propext`, `Classical.choice`, `Quot.sound`: no `sorry` and no additional axioms.

Main statement in Lean. For an instance $(A, a_1, \dots, a_{3m})$ of Three-Partition, the graphs G and H arise from the union of paths on a_i vertices and the union of m cycles on A vertices by attaching M leaves to every vertex. Let $1 \le p \le \infty$. `GsimOT.Hardness.ThreePartition.dOT_le_one_of_isYes`: for a YES-instance with all $a_i \ge 2$, $d_p^{\mathrm{OT}}(G,H) \le 1$. `GsimOT.Hardness.ThreePartition.lt_dOT_of_not_isYes`: for a NO-instance with all $a_i \ge 3$, $0 < \varepsilon \le 1$ and M as in Eq. (46), $d_p^{\mathrm{OT}}(G,H) > 2^{b(p)} - \varepsilon$.

Correspondence

Paper	Page	Statement	Lean declaration(s) and location

Eq. (45)	53	The aggregated matrix $A^Q_{G,H} = A_G - QA_HQ^\top$, $A^Q_{G,H}(v,v') = \sum_{w,w'} Q(v,w)D(v,w,v',w')Q(v',w')$	`GsimOT.Hardness.aggMatrix`
			`Theorem17_OperatorNormHardness/Lemma53_AggregatedMatrix.lean:88`
			`GsimOT.Hardness.aggMatrix_isSymm`
			`Theorem17_OperatorNormHardness/Lemma53_AggregatedMatrix.lean:91`

Lemma 53	53–54	$\|A^Q_{G,H}\|_p \le \|B_p(G,H,Q)\|_p$ for all $Q \in \mathrm{DS}(V,W)$ and $1 \le p \le \infty$	`GsimOT.Hardness.opNorm_aggMatrix_le` `Theorem17_OperatorNormHardness/Lemma53_AggregatedMatrix.lean:100`

Corollary 54	54	If a column of $A^Q_{G,H}$ has k entries of absolute value $\ge \varepsilon$, then $\|A^Q_{G,H}\|_p \ge k^{b(p)}\varepsilon$, so $\|B_p(G,H,Q)\|_p \ge k^{b(p)}\varepsilon$ by Lemma 53; hence $\|B_p(G,H,Q)\|_p > c$ if $k > (c/\varepsilon)^{1/b(p)}$	`GsimOT.Hardness.rpow_bExp_mul_le_opNorm`
			`Theorem17_OperatorNormHardness/OperatorPNorms.lean:147`
			`GsimOT.Hardness.rpow_bExp_mul_le_opNorm_Bpq`
			`Theorem17_OperatorNormHardness/Corollary54_ColumnBound.lean:41`
			`GsimOT.Hardness.lt_opNorm_Bpq`
			`Theorem17_OperatorNormHardness/Corollary54_ColumnBound.lean:47`

Paper	Page	Statement	Lean declaration(s) and location
Proof of Thm. 17: G, H	54	Paths on a_i vertices and m cycles on A vertices, with M leaves attached to every vertex	`GsimOT.Hardness.ThreePartition.graphG` Theorem17_OperatorNormHardness/Theorem17.lean:103 `GsimOT.Hardness.ThreePartition.graphH` Theorem17_OperatorNormHardness/Theorem17.lean:107
Proof of Thm. 17: observation	54	If $xx' \in E(G)$, then $A^Q_{G,H}(x,x') = \sum_{yy' \notin E(H)} Q(x,y)Q(x',y')$ (and the analogue for non-edges)	`GsimOT.Hardness.aggMatrix_of_adj` Theorem17_OperatorNormHardness/Claim55_LeafMass.lean:23 `GsimOT.Hardness.aggMatrix_of_not_adj` Theorem17_OperatorNormHardness/Claim55_LeafMass.lean:29
Claim 55	55	If $Q(v, L_H) \geq \eta > 0$ for a base vertex v and $M > 2(4/\eta)^2 + 1$, then $\|B_p(G,H,Q)\|_p > 2^{b(p)}$	`GsimOT.Hardness.claim55` Theorem17_OperatorNormHardness/Claim55_LeafMass.lean:71
Claim 56	55–56	If $Q(v,w), Q(v,w') \geq \varepsilon > 0$ for distinct base vertices w, w' (with at most one common neighbour, which follows from $A \geq 5$) and $M > 2(2/\varepsilon)^2 + 1$, then $\|B_p(G,H,Q)\|_p > 2^{b(p)}$	`GsimOT.Hardness.claim56` Theorem17_OperatorNormHardness/Claim56_SplitMass.lean:32
Theorem 17 (YES)	12, 55	YES-instances: $d_p^{\mathrm{OT}}(G,H) \leq 1$ for all $1 \leq p \leq \infty$	`GsimOT.Hardness.ThreePartition.dOT_le_one_of_isYes` Theorem17_OperatorNormHardness/Theorem17.lean:113 `GsimOT.Hardness.opNorm_Bpq_permMat_le_one` Theorem17_OperatorNormHardness/Theorem17.lean:56
Theorem 17 (NO), Eq. (46)	12, 56–57	NO-instances with all $a_i \geq 3$: $d_p^{\mathrm{OT}}(G,H) > 2^{b(p)} - \varepsilon$ for $0 < \varepsilon \leq 1$ and M as in Eq. (46)	`GsimOT.Hardness.ThreePartition.lt_dOT_of_not_isYes` Theorem17_OperatorNormHardness/Theorem17.lean:168
Theorem 17 (w.l.o.g. $a_i \geq 3$)	54	Both bounds after adding 3 to each a_i and 9 to A	`GsimOT.Hardness.ThreePartition.dOT_shift_gap` Theorem17_OperatorNormHardness/Theorem17.lean:345
Theorem 4 (2)	4	Hardness of approximating d_p^{OT} within $2^{b(p)} - \varepsilon$ (complexity side not formalised)	`GsimOT.Hardness.ThreePartition.dOT_shift_gap` Theorem17_OperatorNormHardness/Theorem17.lean:345

Shared notions used (from `GsimOT/Common`)

Paper	Page	Statement	Lean declaration(s) and location

Appendix A	14	ℓ^p norm $\|x\|_p$ $(1 \le p \le \infty)$	`GsimOT.lpNorm`
			`Common/OperatorNorm.lean:25`
			`GsimOT.lpNorm_eq_sum`
			`Common/OperatorNorm.lean:57`
			`GsimOT.lpNorm_top`
			`Common/OperatorNorm.lean:62`

Appendix A	14	Operator norm $\|A\|_{p\to q} = \max_{\|x\|_p=1} \|Ax\|_q$ (in Lean the supremum over the unit ball $\|x\|_p \le 1$, the same value for nonempty index sets)	`GsimOT.opNorm`
			`Common/OperatorNorm.lean:28`
			`GsimOT.lpNorm_mulVec_le`
			`Common/OperatorNorm.lean:128`
			`GsimOT.opNorm_le_of_bound`
			`Common/OperatorNorm.lean:135`
			`GsimOT.exists_lpNorm_mulVec_eq_opNorm`
			`Common/OperatorNorm.lean:142`

Appendix A	14	Hölder conjugate q^* with $1/q^* = 1 - 1/q$	`GsimOT.one_le_conjExponent`
			`Common/OperatorNorm.lean:33`
			`GsimOT.inv_conjExponent_toReal`
			`Common/OperatorNorm.lean:38`

Appendix A	14	Hölder's inequality $\|y^\top z\| \le \|y\|_{q^*}\|z\|_q$ and its equality case	`GsimOT.abs_dotProduct_le_lpNorm_mul`
			`Common/OperatorNorm.lean:188`
			`GsimOT.exists_dotProduct_eq_lpNorm`
			`Common/OperatorNorm.lean:224`

Lemma 19	14	$\|A\|_{p\to q} = \max_{\|y\|_{q^*}=1,\|x\|_p=1} \|y^\top Ax\|$ (in Lean over the unit balls)	`GsimOT.lemma19`
			`Common/OperatorNorm.lean:290`
			`GsimOT.abs_dotProduct_mulVec_le_opNorm`
			`Common/OperatorNorm.lean:267`
			`GsimOT.exists_abs_dotProduct_mulVec_eq_opNorm`
			`Common/OperatorNorm.lean:281`

Paper	Page	Statement	Lean declaration(s) and location

–	–	Triangle inequality, $\|A\|_{p\to q} \le \sum_{i,j} \|A_{ij}\|$ and continuity of $\|\cdot\|_{p\to q}$	`GsimOT.opNorm_add_le`
			`Common/OperatorNorm.lean:169`
			`GsimOT.opNorm_le_sum_abs`
			`Common/OperatorNorm.lean:158`
			`GsimOT.abs_opNorm_sub_le`
			`Common/Attainment.lean:78`
			`GsimOT.continuous_opNorm`
			`Common/Attainment.lean:91`

Appendix A	15	Restricting to rows and columns "cannot increase the operator norm"	`GsimOT.lpNorm_le_of_abs_le`
			`Common/Attainment.lean:39`
			`GsimOT.opNorm_restrict_le`
			`Common/Attainment.lean:53`

Section 1, Appendix A	2–3, 14	Couplings $\mathrm{DS}^*(V,W)$; x^a with $0^a := 0$ (convention $0^0 = 0$)	`GsimOT.IsCoupling`
			`Common/OTDistances.lean:36`
			`GsimOT.powZ`
			`Common/OTDistances.lean:42`

Section 1, Eqs. (1), (2)	2–3	$B(G,H,Q)$ and $B_{p\to q}(G,H,Q)$	`GsimOT.B`
			`Common/OTDistances.lean:58`
			`GsimOT.Bmat`
			`Common/OTDistances.lean:70`

Eq. (2), Section 2	3, 5	$\delta^{\mathrm{OT}}_{p\to q}$ (couplings) and $d^{\mathrm{OT}}_{p\to q}$ ($\mathrm{DS}(V,W)$), both infima	`GsimOT.deltaOTPQ`
			`Common/OTDistances.lean:74`
			`GsimOT.dOT`
			`Common/OTDistances.lean:79`

Sections 1–2, Appendices A–B	3, 5, 14–15	$\delta^{\mathrm{OT}}_{p\to q} = n^{-\kappa_{p\to q}} d^{\mathrm{OT}}_{p\to q}$ for graphs of the same order n ($d^{\mathrm{OT}}_{p\to q}$ is the unnormalised version, $s(n) = n^{\kappa_{p\to q}}$), via $\mathrm{DS}(V,W) = n\,\mathrm{DS}^*(V,W)$ and the homogeneity of $B_{p\to q}$ and of $\|\cdot\|_{p\to q}$	`GsimOT.deltaOTPQ_eq_dOT`
			`Common/OTDistances.lean:131`
			`GsimOT.Bmat_smul`
			`Common/OTDistances.lean:117`
			`GsimOT.opNorm_smul_of_nonneg`
			`Common/OTDistances.lean:94`

Paper	Page	Statement	Lean declaration(s) and location
Appendix A	14–15	"All minima … are attained" for $\delta^{\mathrm{OT}}_{p\to q}$ and $d^{\mathrm{OT}}_{p\to q}$: lower semicontinuity of $Q \mapsto \|B_{p\to q}(G,H,Q)\|$, compactness	`GsimOT.lowerSemicontinuous_opNorm_Bmat` Common/Attainment.lean:136 `GsimOT.isCompact_setOf_isCoupling` Common/Attainment.lean:168 `GsimOT.exists_deltaOTPQ_eq` Common/Attainment.lean:199 `GsimOT.exists_dOT_eq` Common/Attainment.lean:223
Proofs of Thms. 15–17	50–57	Disjoint unions of paths (F_1) and of cycles; their degrees and edge counts	`GsimOT.Hardness.pathUnion` Common/PathsAndCycles.lean:56 `GsimOT.Hardness.cycleUnion` Common/PathsAndCycles.lean:226 `GsimOT.Hardness.sum_adj_pathUnion` Common/PathsAndCycles.lean:134 `GsimOT.Hardness.deg_cycleUnion` Common/PathsAndCycles.lean:238
Proof of Thm. 15	50	Three-Partition instances $(A, a_1, \ldots, a_{3m})$; the step of adding 3 to each a_i and 9 to A	`GsimOT.Hardness.ThreePartition` Common/ThreePartition.lean:26 `GsimOT.Hardness.ThreePartition.IsYes` Common/ThreePartition.lean:39 `GsimOT.Hardness.ThreePartition.shift` Common/ThreePartition.lean:206 `GsimOT.Hardness.ThreePartition.shift_isYes_iff` Common/ThreePartition.lean:223
Proofs of Thms. 15–17	50–56	A solution gives a bijection mapping each path into one component; conversely a $Q \in \mathrm{DS}$ mapping every path into a single component gives a solution (counting mass)	`GsimOT.Hardness.ThreePartition.layoutEquiv` Common/ThreePartition.lean:137 `GsimOT.Hardness.ThreePartition.isYes_of_isDS` Common/ThreePartition.lean:168
Proofs of Thms. 17, 18	54, 57	Attaching $\ell(b)$ leaves to every vertex b of a base graph	`GsimOT.Hardness.addLeaves` Common/Leaves.lean:33

Notes on the formalisation

1. Only the correctness of the reduction is formalised: for the constructed graphs, YES-instances have a small and NO-instances a large distance. The complexity side (polynomial running time, NP-hardness of Three-Partition, Håstad's bound for MaxCut) is not formalised. The results of Section 4 and their shared gadgets are in the namespace `GsimOT.Hardness`.

2. Theorem 17 is proved with the bound $2^{b(p)} - \varepsilon$ of its statement. The paper fixes $0 < \varepsilon^* < \sqrt{2} - 1$ (p. 54, l. 1728), so that $2^{b(p)} - \varepsilon^* > 1$ (l. 1729–1730); the Lean statement holds for all $0 < \varepsilon \leq 1$, which includes this range.

3. Lemma 53 is proved without the duality of Lemma 19 (Appendix A): $A^Q x$ is the Q-average of $B_p \tilde{x}$ with $\tilde{x}(v, w) = Q(v, w)^{1/p} x(v)$ (the paper's $\tilde{x}$), and the averaging is an ℓ^p contraction by Jensen. Corollary 54 uses the test vectors e_v and a normalised sign vector on the column support instead of $\|A\|_p = \|A^\top\|_{p^*}$.

4. YES-instances: the paper observes that $A_H - A_G^\pi$ is the adjacency matrix of $3m$ disjoint edges, so $\|A_G^\pi - A_H\|_p = 1$ (p. 55, l. 1736–1737); the formalisation bounds it by 1 by the Schur test (all absolute row and column sums are at most 1).

5. Claim 56 takes $|N_H(w) \cap N_H(w')| \leq 1$ as a hypothesis; it is discharged with $A \geq 7$ in the proof of the theorem.

6. Remark 62 (other operator $p \to q$-norms, p. 62) is not formalised.